

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : All Semester :I and II Subject Code & Name: BTBS102P (Engineering Physics) Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	What are free oscillations? Obtain the differential equation of free oscillations.	CO1	06
B)	Define resonance and explain sharpness of resonance with suitable graph.	CO1	06
C)	Define ultrasonic waves. Write any two applications of ultrasonic. Calculate natural frequency of 40 mm length of pure iron by using the magnetostriction oscillator to produce ultrasonic waves. Given: Young's Modulus is $115 \times 10^9 \text{ N/m}^2$ Density of pure iron is $7.25 \times 10^3 \text{ Kg/m}^3$	CO1	06
Q.2	Solve Any Two of the following.		
A)	Discuss Interference in thin film due to reflected light.	CO2	06
B)	Distinguish between spontaneous and stimulated emission of radiation. Explain Huygen's theory of double refraction.	CO2	06
C)	What is principle of fibre optics? Define acceptance angle and numerical aperture. In an optical fiber, the core material has refractive index 1.48 & refractive index of clad material is 1.4. Find numerical aperture.	CO2	06
Q. 3	Solve Any Two of the following.		
A)	Explain motion of electron in parallel and perpendicular electric field	CO3	06
B)	With the help of graph and suitable diagram discuss GM counter.	CO3	06
C)	Give physical significance of wave function ψ . Derive Schrodinger's time dependent wave equation.	CO3	06
Q.4	Solve Any Two of the following.		
A)	Define Packing Density. Find Packing Density for SC, BCC and FCC	CO4	06
B)	Explain continuous X-ray spectra in detail. A Coolidge tube operates at 50 kv. Find minimum wavelength of generated X-rays.	CO4	06
C)	Write Maxwell's equations in Integral and Differential form.	CO4	06
Q. 5	Solve the following.		
A)	What is magnetic domain? Discuss the different types of magnetic materials on the basis of magnetic moment.	CO5	06
B)	Explain phenomenon of superconductivity with suitable graph. The critical field for Niobium is $1 \times 10^5 \text{ A/m}$ at 8K & $2 \times 10^5 \text{ A/m}$ at 0K. Calculate the transition temperature of the element.	CO5	06
*** End ***			